Electromagnetism and Light II

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1 Lessons

1.1 Electromagnetism and light II: currents, AC, and optics

Lesson 1 built the statics (charges, fields, potential, capacitors, DC circuits), the Lorentz force, induction, and the wave idea. This lesson finishes the Finnish high-school picture: currents as *sources* of magnetic fields, magnetism in matter, AC and the transformer, the electromagnetic spectrum, and the whole optics half of FY7 (reflection, refraction, interference, diffraction, polarization). Bits marked *Extra* go beyond the exams but are one step away.

Route:

1. magnetism in matter: dipoles, domains, ferromagnetism;
2. magnetic fields of currents: straight wire, parallel wires, solenoid;
3. charged particles in combined fields: selector, mass spectrometer, accelerators;
4. eddy currents (and self-induction as an extra);
5. alternating current and effective values;
6. the transformer and power transmission;

7. the spectrum, black-body radiation, photon energies;
8. geometric optics: reflection, refraction, total internal reflection, lenses;
9. wave optics: double slit, grating, single-slit diffraction, thin films;
10. polarization;
11. how antennas radiate (extra);
12. common mistakes.

Constants (lesson 1 plus): $\mu_0 = 4\pi \times 10^{-7}$ T m/A, $h = 6.63 \times 10^{-34}$ J s, $\sigma = 5.67 \times 10^{-8}$ W/(m^2 K^4), Wien's $b = 2.90 \times 10^{-3}$ m K, $u = 1.66 \times 10^{-27}$ kg, Earth's field ≈ 50 μT.

1.1.1 Magnetism in matter

Definition 1.1 (Magnetic dipole)**.** A small current loop, a compass needle, and (at the classical level) an orbiting electron all act as a *magnetic dipole*: far away their field has the same shape, and in an external field $\vec{B}$ the dipole twists to align with the field, like a compass needle. The strength is the dipole moment $\vec{\mu}$ (loop: $\mu = IA$, direction by the right-hand rule).

Most magnetism in matter comes from electron spin plus orbital motion, but the organizing fact for FY7 is simpler: materials differ in how their atomic dipoles respond.

- *Ferromagnetic* (iron, nickel, cobalt): neighbouring dipoles lock parallel in regions called *domains*. An unmagnetized nail has randomly oriented domains that cancel; an external field grows the aligned domains at the expense of others, and some alignment stays when the field is removed—a permanent magnet. Heating above the *Curie temperature* (iron: 770 °C) shakes the alignment apart.
- *Paramagnetic* (aluminium, oxygen): dipoles align weakly with the field, Randall style—pulled in slightly, no memory.
- *Diamagnetic* (copper, water, bismuth): dipoles oppose weakly—pushed out slightly. All matter is slightly diamagnetic; it is just usually drowned out.

Remark (Hysteresis, qualitatively). Magnetizing iron is not reversible: ramping the external field up and back down traces a loop (*hysteresis*), because domain walls snag. The loop area is energy lost as heat per cycle—which is why transformer cores use "soft" iron with a narrow loop, and permanent magnets use "hard" steel with a wide one. Plotting the loop is a standard lab exercise, not an exam derivation.

1.1.2 Magnetic fields of currents

Lesson 1 only asked what a field *does* to charges. Now the reverse: currents *make* fields.

Theorem 1.2 (Field of a long straight wire)**.** *A long straight wire carrying current I produces at perpendicular distance r the field*

$$B = \frac{\mu_0 I}{2\pi r}, \tag{1}$$

circling the wire with direction given by the right-hand rule: thumb along the current, fingers curl along $\vec{B}$.

Example 1.3 (How big is it?). $I = 10$ A at $r = 2.0$ cm:

$$B = \frac{4\pi \times 10^{-7} \cdot 10}{2\pi \cdot 0.020} = 1.0 \times 10^{-4}\ \text{T} = 0.10\ \text{mT},$$

about twice Earth's field. Household currents magnetically matter only centimetres away—which is why a clamp meter must hug the wire.

Theorem 1.4 (Force between parallel currents). *Two long parallel wires distance d apart, carrying I_1 and I_2, exert on each other over length L*

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}. \tag{2}$$

Currents in the same *direction attract, opposite directions repel.*

Reason in one line: wire 1's field at wire 2 is $B = \mu_0 I_1/(2\pi d)$ by eq. (1), and wire 2, sitting in that field, feels $F = I_2 L B$ by the wire form of the Lorentz force. Direction: same-direction currents give opposite-facing "gears" of field lines that pull together—check it with the right-hand rule once, then remember the slogan.

Remark (Extra: the old ampere). Until 2019 the ampere was *defined* by eq. (2): 1 A is the current that gives 2×10^{-7} N/m between wires 1 m apart. Since 2019 the ampere is defined by fixing e, and the force is something to measure. Exam-wise nothing changes; it is just nice to know the 2×10^{-7} is not an accident.

Theorem 1.5 (Solenoid). *A long coil (solenoid) of N turns over length ℓ carrying I has an approximately uniform interior field*

$$B = \mu_0 \frac{N}{\ell} I, \tag{3}$$

with field lines running along the axis (right-hand rule: curl fingers with the current, thumb points to the "north" end). Outside, the field is weak—a solenoid is the magnetic analogue of the parallel-plate capacitor.

Example 1.6 (Lab solenoid). $N = 500$ turns over $\ell = 25$ cm with $I = 2.0$ A:

$$B = 4\pi \times 10^{-7} \cdot \frac{500}{0.25} \cdot 2.0 \approx 5.0 \times 10^{-3}\ \text{T} = 5.0\ \text{mT},$$

a hundred times Earth's field—enough to visibly deflect a compass across the bench. Slip an iron core in and the domains of theorem 1.1 multiply the field by hundreds: an electromagnet.

1.1.3 Particles in combined fields, and accelerators

Proposition 1.7 (Velocity selector). *Send a beam through perpendicular uniform $\vec{E}$ and $\vec{B}$ with $\vec{v} \perp \vec{E}, \vec{B}$. The electric force qE and magnetic force qvB cancel exactly when*

$$v = \frac{E}{B}, \tag{4}$$

independent of q and m. Slower particles bend toward the electric side, faster ones toward the magnetic side—so a slit downstream passes only speed E/B.

Proof. Take $\vec{E}$ along $+y$, $\vec{B}$ along $+z$, $\vec{v}$ along $+x$. Then $\vec{F}_E = qE\,\hat{y}$ and $\vec{F}_B = q\vec{v} \times \vec{B} = -qvB\,\hat{y}$. Zero net force needs $qE = qvB$, and q cancels. □

Feed the selected beam into a pure field B' and $r = mv/(qB')$ separates masses: the *mass spectrometer*. With v fixed by eq. (4), $r \propto m/q$, so isotopes land at different spots—this is exercise 2.2.

Remark (Accelerators: the FY7 minimum). A *cyclotron* is lesson 1's circle plus a kick: two half-discs ("dees") in a uniform B with an alternating voltage between them, timed to the lap frequency

$$f = \frac{qB}{2\pi m},$$

which is independent of v (and hence of radius) as long as $v \ll c$—each lap adds the same energy. Example: protons in $B = 1.0$ T give $f \approx 15$ MHz. A *linac* is a straight line of drift tubes with alternating voltages; a *synchrotron* ramps B up as the particles gain (relativistic) momentum so the orbit radius stays fixed. All three steer with magnetism and accelerate with electric fields—magnetism bends, electricity pays, because $\vec{B}$ does no work.

1.1.4 Eddy currents

Definition 1.8 (Eddy currents)**.** A changing flux through a *bulk* conductor induces loop currents inside the metal itself—*eddy currents* (pyörrëvirrat). By Lenz's law they oppose the change: a magnet dropped through a copper tube falls slowly, and an aluminium disc swinging between magnet poles brakes hard (a standard demo and *eddy-current brake*).

Eddies are Lenz's law with no coil to point at, and they cut both ways: induction hobs and induction furnaces cook with them on purpose, while transformer cores laminate the iron into insulated sheets so the loops cannot close—thin sheets, thin losses. Whenever an exam asks "why laminated?", the answer is these two words.

Remark (Extra: self-induction). A coil's own changing current changes its own flux, so it induces a back-voltage opposing the change, $U = -L\,\Delta I/\Delta t$, with inductance L in henries. Sparks when unplugging an inductive load, and the delayed current rise in a coil, are this. FY7 never asks for L quantitatively.

1.1.5 Alternating current

Rotate the generator coil of lesson 1 at angular frequency $\omega = 2\pi f$ and the flux is $\Phi = BA\cos(\omega t)$, so Faraday gives a sinusoidal voltage

$$u(t) = U_{\max}\sin(\omega t), \qquad i(t) = I_{\max}\sin(\omega t) \tag{5}$$

(the resistive case: current in step with voltage). Finland's mains: $f = 50$ Hz ($T = 20$ ms, $\omega = 314$ rad/s).

Proposition 1.9 (Effective (rms) values)**.** *A sinusoidal voltage with peak U_{max} heats a resistor exactly like a DC voltage of*

$$U_{rms} = \frac{U_{max}}{\sqrt{2}}, \qquad I_{rms} = \frac{I_{max}}{\sqrt{2}}, \tag{6}$$

and the average heating power is $P = U_{rms}I_{rms} = U_{rms}^2/R$.

Proof sketch. Instantaneous power is $p = u^2/R = U_{\max}^2\sin^2(\omega t)/R$. Over a full cycle $\sin^2$ averages to $1/2$ (write $\sin^2 = (1-\cos 2\omega t)/2$; the cosine averages away). So $\bar{P} = U_{\max}^2/(2R) = (U_{\max}/\sqrt{2})^2/R$. □

Socket "230 V" means $U_{\rm rms} = 230$ V; the peak is $230\sqrt{2} \approx 325$ V, and the insulation must survive that, not 230. Multimeters read rms by default—a fact worth one exam point some year.

Example 1.10 (Heater on mains)**.** $R = 48\ \Omega$ across 230 V rms: $I_{\rm rms} = 230/48 = 4.8$ A,

$$P = \frac{230^2}{48} \approx 1.1\ \text{kW},$$

and one hour eats 1.1 kWh ≈ 4.0 MJ. The current reverses 100 times per second and the heater never notices, because heating goes as i^2.

1.1.6 The transformer and the grid

Proposition 1.11 (Ideal transformer)**.** *Two coils sharing one iron core, with turns N_1, N_2, satisfy*

$$\frac{U_2}{U_1} = \frac{N_2}{N_1}, \qquad \frac{I_2}{I_1} = \frac{N_1}{N_2} \quad \textit{(ideal, loaded)}, \tag{7}$$

so $U_1 I_1 = U_2 I_2$: power in equals power out.

Proof sketch. The core guides (nearly) all flux through both coils, so each turn sees the same $\Delta\Phi/\Delta t$ and Faraday gives $U_1 = N_1\,\Delta\Phi/\Delta t$, $U_2 = N_2\,\Delta\Phi/\Delta t$—divide and the flux cancels. The current ratio follows from energy conservation: what the secondary delivers, the primary must draw. Real transformers lose a few percent to resistance, hysteresis (section 1.1.1), and eddy currents (theorem 1.8). exercise 2.7 asks for the careful version. □

Why AC won: voltage transforms, and high voltage is how power travels. For delivered power $P = UI$, the current is $I = P/U$ and the line's own resistance burns $P_{\text{loss}} = RI^2 = RP^2/U^2$—doubling the voltage quarters the loss. So the grid steps up to 110–400 kV for transport and back down to 230 V at your socket. exercise 2.3 runs the numbers that make DC transmission lose by two orders of magnitude. Note the transformer needs *changing* flux, which is why the grid is AC at all.

1.1.7 The spectrum and black-body radiation

An accelerating charge radiates (section 1.1.11 below); the frequency equals the charge's oscillation frequency, and $c = \lambda f$ stretches the whole family—radio to gamma—along one line:

band	λ range	typical source
radio	mm–km	antennas, phones, FM ($\sim$ 100 MHz)
microwave	mm–cm	radar, Wi-Fi, Gunn diode
infrared	μm–mm	warm bodies, remote controls
visible	400–700 nm	Sun, LEDs, lasers
ultraviolet	10–400 nm	Sun, mercury lamp
X-ray	10 pm–10 nm	electron tubes, synchrotrons
gamma	$<$ 10 pm	nuclei, cosmic events

Light is one octave of thirty. What differs across bands is what the wave *does*—it penetrates, ionises, or resonates with different-sized things—not what it *is*.

Photon energies (FY8 owns the photon; FY7 needs the numbers): $E = hf = hc/\lambda$. Green 550 nm light carries 3.6×10^{-19} J ≈ 2.3 eV per photon; an FM photon at 100 MHz carries 6.6×10^{-26} J—ten million billion times less, which is why one radio photon does nothing and a mole of them runs the speaker.

Theorem 1.12 (Black-body spectrum: Wien and Stefan–Boltzmann)**.** *A hot object radiates at all wavelengths with a peak at*

$$\lambda_{max}\, T = b = 2.90 \times 10^{-3}\ \text{m K} \qquad \textit{(Wien)}, \tag{8}$$

and total power per area

$$\frac{P}{A} = \sigma T^4 \qquad \textit{(Stefan–Boltzmann)}. \tag{9}$$

Example 1.13 (Two bodies)**.** Sun, $T = 5800$ K: $\lambda_{\max} = 2.90 \times 10^{-3}/5800 \approx 500$ nm—green, mid-visible, which is no coincidence for eyes. Human skin, $T = 310$ K: $\lambda_{\max} \approx 9.3$ μm, deep infrared—night-vision goggles see you by your own glow. And $5800^4/2800^4 \approx 18$: per square metre the Sun's surface out-radiates a 2800 K bulb filament eighteenfold.

1.1.8 Geometric optics

Light travels straight in uniform medium and obeys two laws at boundaries. Angles are always measured from the *normal*.

Theorem 1.14 (Reflection and refraction)**.** Reflection: *angle in equals angle out,* $\theta_1 = \theta_1'$*, in the plane of incidence.* Refraction (Snell):

$$n_1 \sin\theta_1 = n_2 \sin\theta_2, \tag{10}$$

where $n = c/v$ *is the refractive index. Bending is always toward the normal entering higher* n*.*

Example 1.15 (Air to glass)**.** $n_1 = 1.00$, $n_2 = 1.52$, $\theta_1 = 45°$:

$$\sin\theta_2 = \frac{\sin 45°}{1.52} = 0.465 \implies \theta_2 \approx 27.7°.$$

Reverse the ray (glass to air at 27.7°) and it exits at 45°—Snell is symmetric, which the next result exploits.

Proposition 1.16 (Total internal reflection)**.** *Going from higher* n_1 *to lower* n_2*, Snell demands* $\sin\theta_2 = (n_1/n_2)\sin\theta_1 > \sin\theta_1$*. Beyond the* critical angle

$$\sin\theta_c = \frac{n_2}{n_1} \tag{11}$$

no refraction exists: all light reflects. Glass–air: $\sin\theta_c = 1/1.52$*,* $\theta_c \approx 41°$*. Diamond (*$n = 2.42$*) traps everything past* 24°*—that is the sparkle.*

Optical fibre is theorem 1.16 as infrastructure: core $n \approx 1.5$, cladding slightly lower, light launched inside θ_c bounces kilometres with almost no loss, and because frequencies (colours) don't cross-talk, one fibre carries terabits. Endoscopes, broadband, decorative lamps—same law.

Remark (Dispersion)*.* n depends slightly on λ (glass bends blue more than red), so a prism fans white light into a spectrum and lenses fringe in colour unless corrected (achromats). Rainbows are dispersion plus one internal reflection in each droplet.

Proposition 1.17 (Thin lens and flat mirror (standard toolkit))**.** *Flat mirror: image as far behind as object is in front, same size, laterally reversed. Thin lens/converging mirror, real-is-positive convention:*

$$\frac{1}{f} = \frac{1}{a} + \frac{1}{b}, \qquad m = -\frac{b}{a}, \tag{12}$$

a object distance, b image distance, m magnification (negative = inverted). Converging lens: $f > 0$*.*

Example 1.18 (Projector lens)**.** $f = 10$ cm, slide at $a = 15$ cm: $1/b = 1/10 - 1/15 = 1/30$, so $b = 30$ cm and $m = -2$—an inverted image twice life-size, which is why projector slides go in upside down.

1.1.9 Wave optics

Geometric optics fails exactly where waves matter: light plus light can give darkness. The rule is Huygens': every point of a wavefront re-radiates, and the field downstream is the *sum* of all contributions—superposition again, now with phases.

Theorem 1.19 (Double slit and grating)**.** *Coherent light of wavelength λ through slits spaced d apart reinforces in directions*

$$d \sin \theta_m = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \tag{13}$$

(bright fringes; dark halfway between). On a screen at distance $L \gg d$, fringe spacing is $\Delta y = \lambda L/d$. A grating *with hundreds of slits per mm obeys the same eq.* (13) *with far sharper, brighter maxima.*

Proof sketch. At angle θ the second slit's path is longer by $d \sin \theta$ (draw the right triangle at the slits; small-angle geometry). Constructive interference needs that extra path to be a whole number of wavelengths. □

Example 1.20 (Fringes you can measure with a ruler)**.** $d = 0.25$ mm, $L = 1.5$ m, $\lambda = 650$ nm:

$$\Delta y = \frac{6.50 \times 10^{-7} \cdot 1.5}{2.5 \times 10^{-4}} \approx 3.9 \text{ mm}.$$

Measure Δy and you have measured λ—this is exercise 2.6 with a grating, where the sharpness makes the angle precise.

Proposition 1.21 (Single-slit diffraction)**.** *One slit of width a gives* minima *(dark) at*

$$a \sin \theta_m = m\lambda, \qquad m = \pm 1, \pm 2, \ldots \tag{14}$$

so the central bright band has width $2L\lambda/a$ on a screen at L. Narrower slit, wider smear—squeezing light sideways spreads it.

Proof sketch. Pair up points half a slit apart: when $a \sin \theta = \lambda$, each pair differs by half a wavelength and cancels. Repeat per quarter for $m = 2$, and so on. □

Example 1.22 (How wide is "wide"?)**.** $a = 0.10$ mm, $\lambda = 650$ nm, $L = 1.5$ m: first minimum at $\sin \theta = 6.5 \times 10^{-3}$ ($\theta \approx 0.37°$), central band $2 \cdot 1.5 \cdot 6.5 \times 10^{-7}/10^{-4} \approx 2.0$ cm wide. The $m = 0$ exception in eq. (14) is the point: straight through always reinforces.

Remark (Coherence and thin films). eq. (13) needs a fixed phase relation (*coherence*): laser light interferes across the bench; white light only over microns, which is why oil films and soap bubbles show colour—the film is thin enough that front- and back-surface reflections still "remember" each other's phase, and different λ reinforce at different thicknesses. Anti-reflection coatings are the same trick tuned to cancel: thickness $\lambda/(4n)$.

1.1.10 Polarization

Definition 1.23 (Polarization)**.** An EM wave is *transverse*: $\vec{E}$ oscillates perpendicular to travel. *Polarization* is the direction of $\vec{E}$. Unpolarized light (Sun, bulbs) mixes all directions; a polarizer passes one component. This has no sound-wave analogue—air vibrates along travel, so "polarized sound" is meaningless.

Cross two polarizers and nothing gets through (the second eats what the first passed); slip a third at 45° between them and light returns—the middle one rotates the vote, so to speak. Uses: glare-cutting sunglasses (reflected light is partly polarized), LCD screens (liquid crystals twist the polarization pixel by pixel), polarization microscopy, bee navigation by sky polarization.

Remark (Extra: Malus's law). Already-polarized intensity I_0 through a polarizer at angle θ gives $I = I_0 \cos^2 \theta$: project the $\vec{E}$ vector, square for intensity. Three-filter paradox above follows in one line each.

1.1.11 How antennas radiate (extra)

Stationary charges give Coulomb fields; *accelerating* charges also shed a kink that propagates at c—the EM wave. A dipole antenna drives electrons sinusoidally up and down; the radiated $\vec{E}$ is parallel to the antenna, strongest sideways, zero along the axis (charges don't radiate along their own motion). Frequency in equals frequency out: 100 MHz drive makes 3 m waves. Reception is the reverse—the wave's $\vec{E}$ shakes the receiving antenna's electrons, Faraday/Lorentz in action. Radio is lesson 1's induction with the coil stretched into a rod and the flux arriving from kilometres away.

Remark (Most common mistakes). a) Right-hand-rule soup: current-field (grip the wire) vs. force (point fingers along v/I, curl toward B). Two different rules—say which one you are using.

b) Parallel wires: same direction *attracts*. The electricity intuition ("likes repel") misfires because the force is mediated by fields, not charges.

c) rms vs. peak: 230 V is already rms. Dividing by $\sqrt{2}$ again is the classic double discount.

d) Transformer ratios upside down: N sits on the same side as U ($U_2/U_1 = N_2/N_1$), but current crosses over ($I_2/I_1 = N_1/N_2$). Energy conservation (UI in $= UI$ out) rebuilds it if memory fails.

e) Snell: angles from the *normal*, and n pairs with the sine on its own side ($n_1 \sin\theta_1 = n_2 \sin\theta_2$). Most errors are measuring from the surface.

f) Grating vs. single slit zero order: double-slit/grating have a bright $m = 0$ straight through; single-slit "$m = 0$" would be the brightest spot of all, so its minima start at $m = \pm 1$.

g) Coherence amnesia: two bulbs never make fringes, however narrow the slits. No fixed phase, no pattern—only equal intensities added.

Remark (Code links). Three genuine ones. (1) A diffraction grating *is* a Fourier transform in glass: slit positions in, spatial frequencies out, intensities as the power spectrum—exercise 2.8 computes one. (2) Interference is complex addition $I = |\sum A_k e^{i\phi_k}|^2$, exactly phased-array beamforming / antenna code. (3) Lenz's law is negative feedback: opposes the change, like a control loop with gain -1; the exponential $v(t)$ decay of a braked conductor is the same ODE as an RC discharge. Where the analogies break: nature sums infinite continuous paths (Feynman would insist), sampled at λ-scale—no mesh needed, and no aliasing until you simulate it.

2 Exercises

Exercise 2.1. Two long straight parallel wires are 8.0 cm apart. Wire 1 carries $I_1 = 6.0$ A, wire 2 carries $I_2 = 4.0$ A, in the same direction.

(a) Find the magnetic field produced by wire 1 at the position of wire 2 (magnitude and direction, described in words with axes you choose).

(b) Find the force per unit length on wire 2 (magnitude and direction). Is it attractive or repulsive?

(c) A solenoid has $N = 400$ turns over $\ell = 20$ cm and carries $I = 3.0$ A. Find its interior field. How many times Earth's field ($\approx 50\ \mu$T) is this?

Exercise 2.2. A beam of singly ionized carbon isotopes ($q = +e$) with a spread of speeds enters a velocity selector with $E = 4.0 \times 10^4$ V/m and $B = 0.20$ T, then enters a spectrometer chamber with $B' = 0.20$ T perpendicular to the beam. In the chamber each ion traces a half-circle and hits a detector plate; the distance from the entry slit to the impact point is the diameter $2r$.

(a) Show that only ions with one specific speed pass the selector, and compute that speed.

(b) Find the orbit radius in the spectrometer for ^{12}C ($m = 12\,u$).

(c) The second isotope ^{13}C lands 2.1 cm further out along the plate. Verify this separation from the formulas (no full recalculation needed: use scaling, and remember the plate measures diameters).

Exercise 2.3. A power station delivers $P = 100$ MW over a line of resistance $R = 2.0\ \Omega$, first at $U_{\text{rms}} = 20$ kV, then stepped up to 400 kV.

(a) Find the rms current and the power lost as heat in the line in each case.

(b) Give the loss as a percentage of delivered power in each case.

(c) Explain in one or two sentences why the grid uses high voltage for transmission and steps it down near consumers, naming the device that makes this possible.

Exercise 2.4. A laser beam ($\lambda = 532$ nm) strikes an air–glass boundary ($n_{\text{glass}} = 1.52$) at 50° to the normal.

(a) Find the angle of the refracted ray inside the glass.

(b) The refracted ray hits the opposite face of a glass slab (glass to air). The slab faces are parallel. Show that the exiting ray is parallel to the incoming ray, and find the critical angle for the glass–air face. Does total internal reflection occur here?

(c) An optical fibre has core $n = 1.50$ and cladding $n = 1.48$. Find the critical angle at the core–cladding boundary and explain in words why light stays in the fibre.

Exercise 2.5. A converging lens has focal length $f = 12$ cm.

(a) An object stands $a = 30$ cm from the lens. Find the image distance b and magnification m. Is the image real or virtual, upright or inverted?

(b) The object is moved to $a = 8.0$ cm (inside the focal length). Find b and m and describe the image (this is the magnifying-glass case).

(c) A flat mirror stands 2.0 m from a person 1.7 m tall. How tall must the mirror be for the person to see their whole body, and at what height must its top edge sit? (Draw the rays; the answer is independent of the 2.0 m.)

Exercise 2.6. In a grating experiment the screen is $L = 1.20$ m from a grating with 500 lines per mm. A green laser produces first-order maxima 33.1 cm on either side of the central maximum.

(a) Find the slit spacing d and the angle θ_1 of the first maximum.

(b) Use $d \sin\theta_1 = \lambda$ to determine the laser wavelength.

(c) The laser is replaced by white light. Explain in words what pattern appears on the screen and why the centre stays white while higher orders smear into spectra.

Exercise 2.7. An ideal transformer has $N_1 = 200$ primary turns and $N_2 = 20$ secondary turns. The primary is fed $U_{1,\text{rms}} = 230$ V at 50 Hz; the secondary drives a load $R = 4.0\ \Omega$.

(a) Starting from Faraday's law applied to each coil, derive $U_2/U_1 = N_2/N_1$ (state the shared-flux assumption explicitly) and find $U_{2,\text{rms}}$.

(b) Find the secondary current and the power delivered to the load. What primary current does energy conservation predict?

(c) Give two mechanisms that make a real transformer deliver slightly less than this, with one sentence each linking to the physics of this lesson.

Exercise 2.8. A plane wave of wavelength λ passes N identical narrow slits spaced d apart. The complex amplitude in direction θ is the phasor sum

$$A(\theta) = A_0 \sum_{k=0}^{N-1} e^{ik\phi}, \qquad \phi = \frac{2\pi d \sin\theta}{\lambda},$$

and the observed intensity is $I(\theta) = |A(\theta)|^2$.

(a) Show that the principal maxima sit exactly at the grating condition $d\sin\theta = m\lambda$, for any N.

(b) Simulate $I(\theta)$ numerically (e.g. in Python) for $N = 2$ and $N = 8$ over $-90° < \theta < 90°$ with $d = 4\lambda$, and plot both. Count the small secondary maxima between neighbouring principal maxima for each N and state the pattern.

(c) Explain in words why increasing N makes the principal maxima sharper, connecting the phasor picture to the experimental fact that a grating resolves wavelengths far better than a double slit.

3 Solutions

Solution to exercise 2.1. (a) It produces a magnetic field of strength:

$$B = \frac{\mu_0 I}{2\pi r}$$
$$B = \frac{\mu_0 \cdot 6.0A}{2\pi 8.0cm}$$
$$B \approx 1.5 \cdot 10^{-7}$$

Using the right hand rule (point thumb towards current, curl fingers to see induced magnetic field), we can see that the when the current is going towards +y, the induced magnetic field on wire 2 (assuming it is to the right of wire 1) points down. (b) The force between parallel currents is given by (derived from $F = I\overline{L} \times \overline{B}$):

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$
$$F = \frac{\mu_0 \cdot 6.0A \cdot 4.0A \cdot L}{2\pi \cdot 8.0cm}$$
$$\frac{F}{L} = \frac{\mu_0 \cdot 6.0A \cdot 4.0A}{2\pi \cdot 0.08m}$$
$$\frac{F}{L} \approx 6.0 \cdot 10^{-5} N/m$$

(c) The interior field can be calculated by:

$$B = \mu_0 \frac{N}{\ell} I$$
$$B = \mu_0 \frac{400}{0.2m} 3.0A$$
$$B \approx 7.54 \cdot 10^{-3} T$$
$$\frac{7.54 \cdot 10^{-3} T}{5 \cdot 10^{-5}} \approx 1.5 \cdot 10^2$$

So the interior magnetic field is about 150 times stronger that Earth's magnetic field. □

Solution to exercise 2.2. (a) To pass the selector, the electric and magnetic forces have to cancel out. Since the fields are perpendicular and their forces are opposed, we have:

$$\begin{gathered} qvB = qE \\ vB = E \\ v = \frac{E}{B} \\ v = \frac{4.0 \cdot 10^4 V/m}{0.20T} \\ v = 2 \cdot 10^5 m/s \end{gathered}$$

(b) In the spectrometer chamber, the atoms experience a single force F:

$$\begin{gathered} \overline{F} = q\overline{v} \times B \qquad (\overline{v} \perp \overline{B}) \\ |\overline{F}| = qvB \\ ma = qvB \\ a = \frac{qvB}{m} \\ \frac{v^2}{r_1} = \frac{qvB}{m} \\ \frac{v}{r_1} = \frac{qB}{m} \\ r_1 = \frac{mv}{qB} \\ r_1 = \frac{12u \cdot 2 \cdot 10^5 m/s}{e \cdot 0.2T} \\ r_1 \approx 0.125m \end{gathered}$$

(c) Its mass is $13u$, so its radius is:

$$\begin{gathered} r_2 = \frac{13}{12} r_1 \\ d_2 = \frac{26}{12} r_1 \\ d_2 - 2r_1 = \frac{1}{6} r_1 \\ d_2 - d_1 \approx 2.1cm \end{gathered}$$

So the 2.1cm difference checks out. □

Solution to exercise 2.3. The rms current is:

$$\begin{gathered} I_{rms} = \frac{P}{U_{rms}} I_{rms,1} = \frac{100MW}{20kV} \\ I_{rms,1} = 5 \cdot 10^3 \\ I_{rms,2} = \frac{100MW}{400kV} \\ I_{rms,2} = 250A \end{gathered}$$

The power lost in each scenario is given by:

$$P_{loss} = RI^2$$
$$P_{loss,1} = RI_{rms,1}^2$$
$$P_{loss,1} = 2.0\Omega \cdot (5 \cdot 10^3)^2$$
$$P_{loss,1} = 5 \cdot 10^7 W$$
$$\frac{Ploss,1}{P} = 0.5$$
$$P_{loss,2} = RI_{rms,2}^2$$
$$P_{loss,2} = 2.0\Omega \cdot (250A)^2$$
$$P_{loss,2} = 2.0\Omega \cdot (250A)^2$$
$$P_{loss,2} = 6.25 \cdot 10^4 W$$
$$\frac{P_{loss,2}}{P} = 0.000625$$

(c) The power loss scales according to the inverse of the voltage squared. Therefore it is extremely beneficial to use high voltage for power grids, and then step it down near consumers for security reasons. The device that facilitates this is called a transformer. □

Solution to exercise 2.4. (a) Using Snell's law:

$$n_1 sin(\theta_1) = n_2 sin(\theta_2) \qquad (n_1 = 1, n_2 = 1.52, \theta_1 = 50^\circ)$$
$$sin(50^\circ) = 1.52 sin(\theta_2)$$
$$\theta_2 = arcsin(\frac{sin(50^\circ)}{1.52})$$
$$\theta_2 \approx 30.3^\circ$$

(b) For glass to air, we have:

$$n_1 sin(\theta_1) = n_2 sin(\theta_2) \qquad (n_1 = 1, n_2 = 1.52, \theta_2 = arcsin(\frac{sin(50^\circ)}{1.52}))$$
$$sin(\theta_1) = 1.52\frac{sin(50^\circ)}{1.52}$$
$$sin(\theta_1) = sin(50^\circ)$$
$$\theta_1 = 50^\circ$$

Thus it is parallel to the original ray. The critical angle for the glass-air face if when Snell's equation has no solutions:

$$sin(\theta_1) = 1.52 sin(\theta_2)$$
$$1.52 sin(\theta_2) = 1$$
$$sin(\theta_2) = \frac{1}{1.52}$$
$$\theta_2 = arcsin(\frac{1}{1.52})$$
$$\theta_2 \approx= 41.1^\circ$$

So the critical angle is about 41.1° for the glass-air face. Total internal reflection does not occur here, since we calculated θ_2 to be about 30.3°, which is smaller than the critical angle that we

calculated. (c) Using Snell's law again:

$$
\begin{aligned}
n_1 sin(\theta_1) &= n_2 sin(\theta_2) \\
1.5 sin(\theta_1) &= 1.48 sin(\theta_2) \\
\frac{1.5}{1.48} sin(\theta_1) &= sin(\theta_2) \\
\frac{1.5}{1.48} sin(\theta_1) &= 1 \\
sin(\theta_1) &= \frac{1.48}{1.5} \\
\theta_1 &= arcsin(\frac{1.48}{1.5}) \\
\theta_1 &\approx 80.6^\circ
\end{aligned}
$$

The critical angle is 80.6°, so light stays in the fibre when it is at most 10° from the parallel of the wire. □

Solution to exercise 2.5. (a) Let's use the equation for converging mirrors/thin lenses:

$$
\begin{aligned}
\frac{1}{f} = \frac{1}{a} + \frac{1}{b}, \qquad m &= -\frac{b}{a} \\
\frac{1}{12} &= \frac{1}{30} + \frac{1}{b} \\
\frac{b}{12} &= \frac{b}{30} + 1 \\
\frac{b}{6} &= \frac{b}{15} + 2 \\
\frac{b}{2} &= \frac{b}{5} + 6 \\
\frac{5b}{10} &= \frac{2b}{5} + 6 \\
\frac{3b}{10} &= 6 \\
3b &= 60 \\
b &= 20(cm) \\
m &= -\frac{b}{a} \\
m &= -\frac{20}{30} \\
m &= -\frac{2}{3}
\end{aligned}
$$

The image distance is around $20cm$, so the image is real, inverted and shrunk by 1.5. (b)

$$\frac{1}{f} = \frac{1}{a} + \frac{1}{b}, \qquad m = -\frac{b}{a}$$
$$\frac{1}{12} = \frac{1}{8} + \frac{1}{b}$$
$$\frac{b}{12} = \frac{b}{8} + 1$$
$$\frac{b}{6} = \frac{b}{4} + 2$$
$$\frac{2b}{12} = \frac{3b}{12} + 2$$
$$-\frac{b}{12} = 2$$
$$-b = 24$$
$$b = -24(cm)$$
$$m = -\frac{-24}{8}$$
$$m = 3$$

So the image is virtual (on the side of the lens), upright and enlarged by 3.
(c) Its top edge must sit at 1.7m for the person to see themselves and the bottom edge can be at most at $\frac{1.7m}{2} = 0.85m$ from the floor. The length of the mirror would then be 0.85m, which does not depend on the distance from the mirror. □

Solution to exercise 2.6. (a) We know that $d = \frac{1}{500}mm$, and we can calculate θ_1:

$$tan(\theta_1) = \frac{0.331}{1.2}$$
$$\theta_1 = arctan(\frac{0.331}{1.2})$$
$$\theta_1 \approx 15.4^\circ$$

(b) Using the equations for grating:

$$d\sin(\theta_m) = \lambda$$
$$d\frac{0.331}{\sqrt{1.2^2 + 0.331^2}} = \lambda$$
$$\frac{1}{500} \cdot \frac{0.331}{\sqrt{1.2^2 + 0.331^2}}mm = \lambda$$
$$\frac{1}{500} \cdot \frac{0.331}{\sqrt{1.2^2 + 0.331^2}} \cdot 10^{-3}m = \lambda$$
$$\lambda \approx 532nm$$

(c) White light is composed of a combination of all wavelengths, so we no longer see the same interference patterns, since there is no more visible constructive/destructive interference. Instead, since longer wavelengths spread further on the screen, we see a prism-like image, where there is white in the middle, then a gradient of hues on either side ending with red light on the extremities. □

Solution to exercise 2.7. (a) Using Faraday's law for induction:

$$U_{ind} = -N\frac{d}{dt}\Phi(t)$$
$$U_1 = -N_1\frac{d}{dt}\Phi_1(t)$$
$$U_2 = -N_2\frac{d}{dt}\Phi_2(t)$$
$$\frac{U_2}{U_1} = \frac{-N_2\frac{d}{dt}\Phi_2(t)}{-N_1\frac{d}{dt}\Phi_1(t)}$$
$$\frac{U_2}{U_1} = \frac{N_2}{N_1}$$

In the final step we assume that $\Phi_1(t) = \Phi_2(t)$, i.e. they share the same magnetic flux. Let's find $U_{2,rms}$:

$$\frac{U_{2,rms}}{U_{1,rms}} = \frac{N_2}{N_1}U_{2,rms} = \frac{N_2}{N_2}\cdot U_{1,rms}U_{2,rms} = \frac{20}{200}\cdot 230VU_{2,rms} = 23V$$

(b) Using Ohm's law:

$$U = RI$$
$$U_{2,rms} = R_2I_2$$
$$I_2 = \frac{U_{2,rms}}{R_2}$$
$$I_2 = \frac{23V}{4.0\Omega}$$
$$I_2 = 5.75A$$

Let's calculate the power:

$$P = UI$$
$$P_2 = U_{2,rms}I_2$$
$$P_2 = 23V\cdot 5.75A$$
$$P_2 = 132.25W$$

If energy were perfectly conserved, we would have $P_1 = P_2$, so:

$$P_1 = P_2$$
$$U_{1,rms}I_1 = U_{2,rms}I_2$$
$$230V\cdot I_1 = 23V\cdot 5.75A$$
$$I_1 = 0.575A$$

(c) The first mechanism is the power dissipated by the wire itself, as we have seen in exercise 2.3. The second mechanism is Eddy currents, which we also saw in the lesson. □

Solution to exercise 2.8. (a) NOTE to self: sum is maximised when $\phi = 2\pi m$, (far right end of the circle), then the equation simplifies to $dsin(\theta) = m\lambda$

□

4 Feedback